Beautiful Real Estate
Multimodal Transformers and Property Values

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“Seeing is believing.”

“One look is worth a thousand words.”

“百闻不如一见”
[To see for oneself is a hundred times better than hearing from others.]

« Un bon croquis vaut mieux qu’un long discours »
[A good sketch is worth more than a long speech]

- A single visual can convey facts, complex ideas, or emotions more effectively than a lengthy verbal description
- Multiple examples of image-based communication
  - Catalogues
  - Menus
  - Murals and stained glass in public buildings and churches

This Paper

- New technique to extract an *image-based* measure of the state of disrepair (alt. “luxuriousness”) of real estate properties

  CLIP-score

- Unlike measures derived from querying LLMs
  - Fully reproducible
  - Relatively efficient computationally
  - Almost completely free
- We demonstrate the method using photos from a large sample of listed Moscow real estate

Findings

1. CLIP-score follows predictable patterns:
   - Positively correlated with self-reported ‘state of repair’
   - U-shaped pattern of real estate quality relative to year of construction
   - The best real estate is in the city center, north-western and western districts
   - Properties is the far east and far south districts is in terrible state of disrepair
2. CLIP-score is a robust predictor of market outcomes
   - Better-looking properties listed for significantly higher prices
   - They also sell/rent faster

Images in Economics Research

Literature

- News Media Photos (Obaid and Pukthuanthong, 2022, Ash et al., 2021)
- Paintings (Gorin et al., 2025)
- Faces (Voth and Yanagizawa-Drott, 2024)
- Google StreetView (Li, 2021)
- Satellite Imagery (Jean et al., 2016)

Contributions

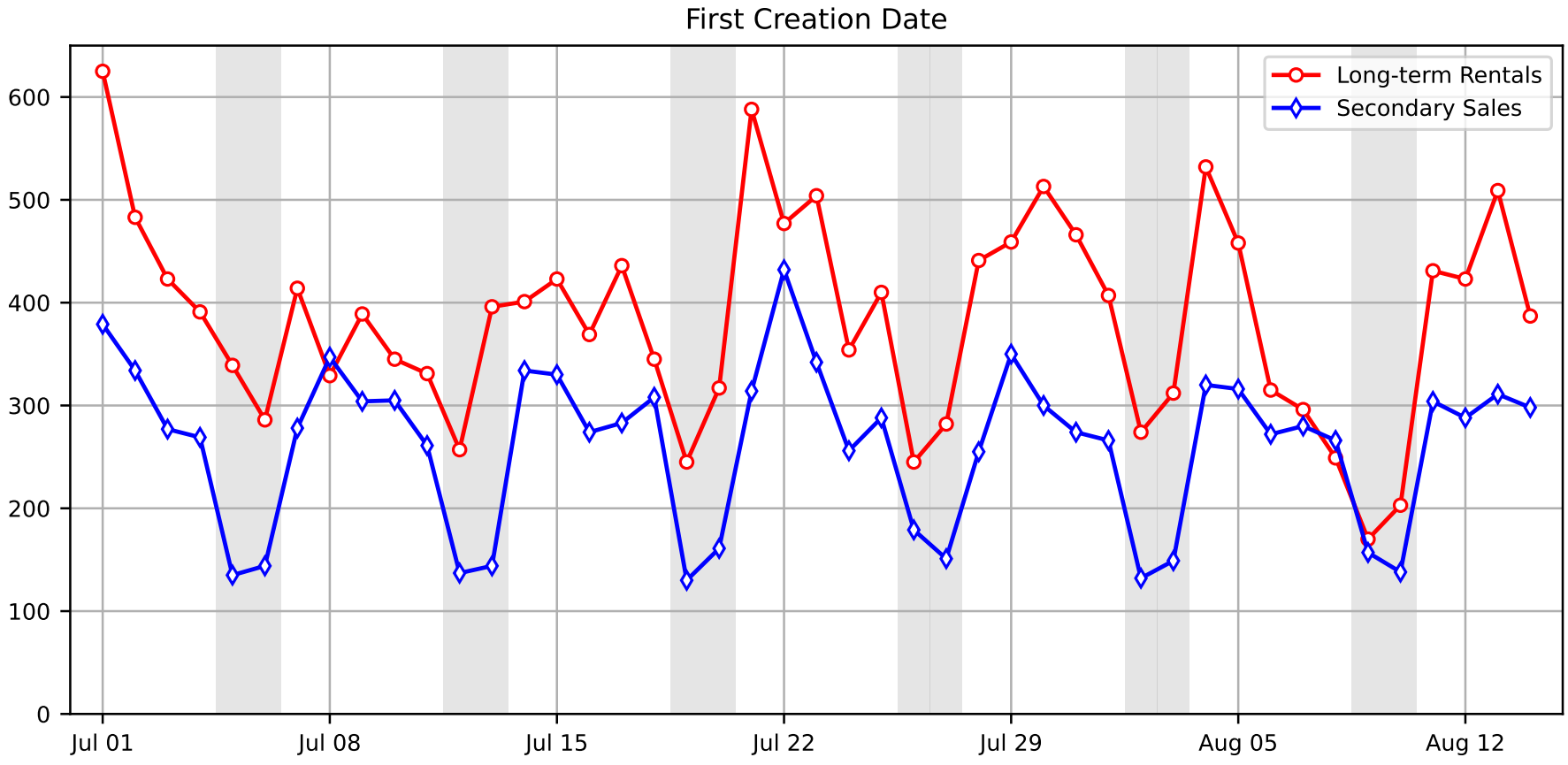
- CLIP
- Real Estate Images

1. Introduction
2. Data
3. Methods
4. Comparing Approaches
   - Replication Study
5. Descriptive Analysis
6. Hedonic Model
   - Gradient Boosting
   - Shapley Values
7. Time on the Market
8. Conclusions

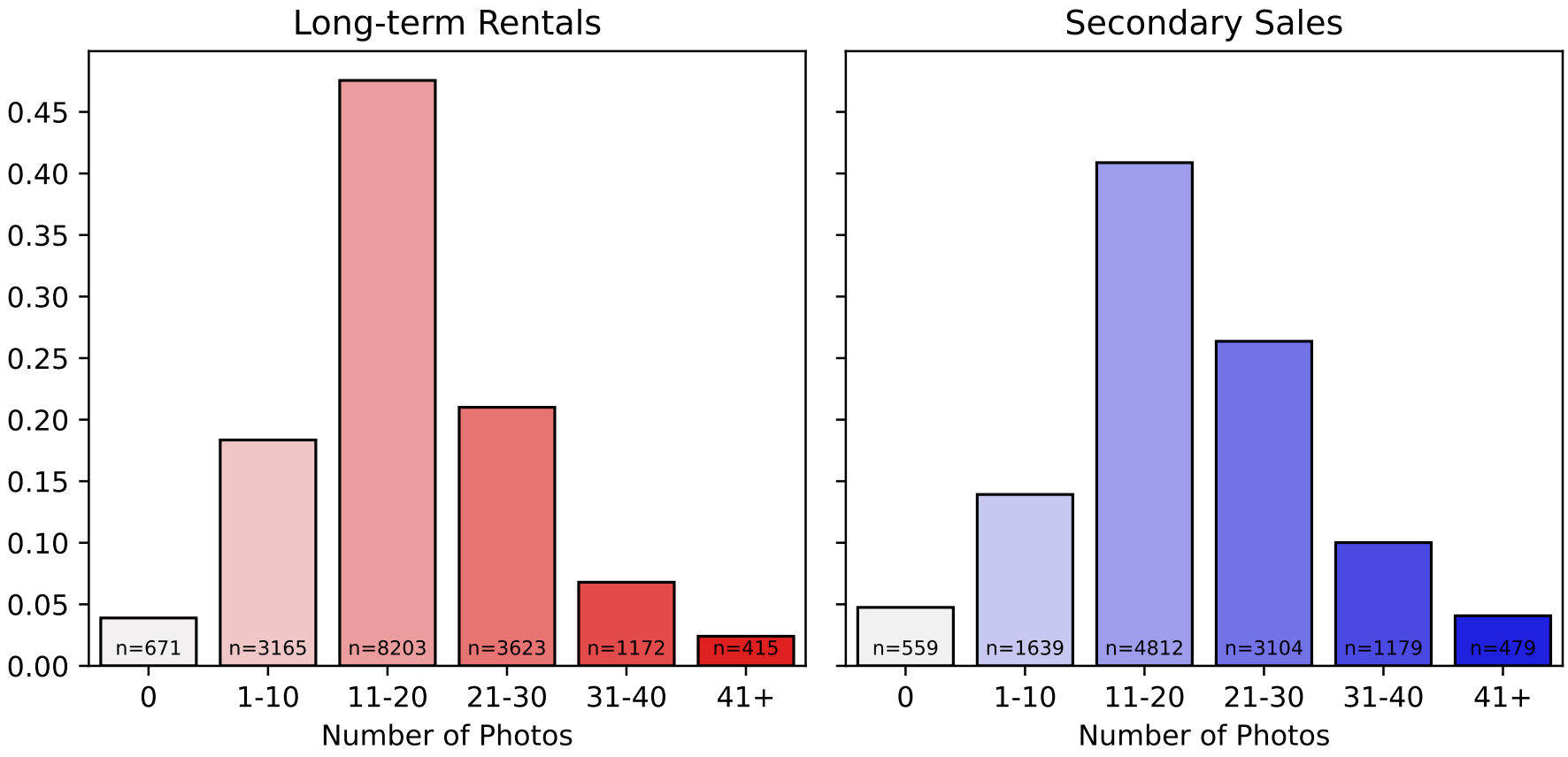
Data Collection

- We crawled `cian.ru` for Moscow properties advertised for long-term rent or secondary sale
- Ads created between July 1st and August 14th 2025
- Data collected:
  - Property: area, rooms/bathrooms, ceiling height, state of repair, floor number
  - Building: coordinates, construction material, age, up for demolition
  - Complete asking price history
  - All photos
- Posts were followed periodically until December 5th

Data Collection Results



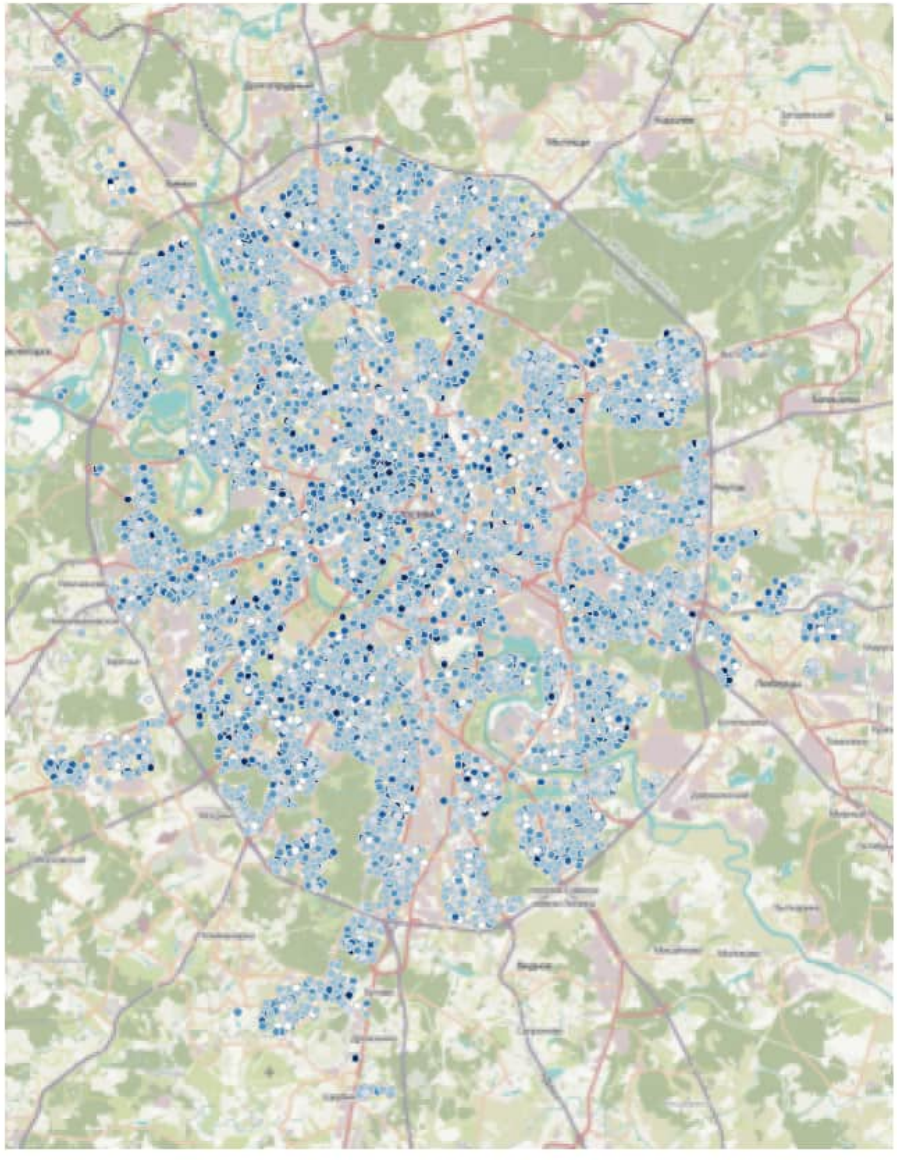
Data Collection Results



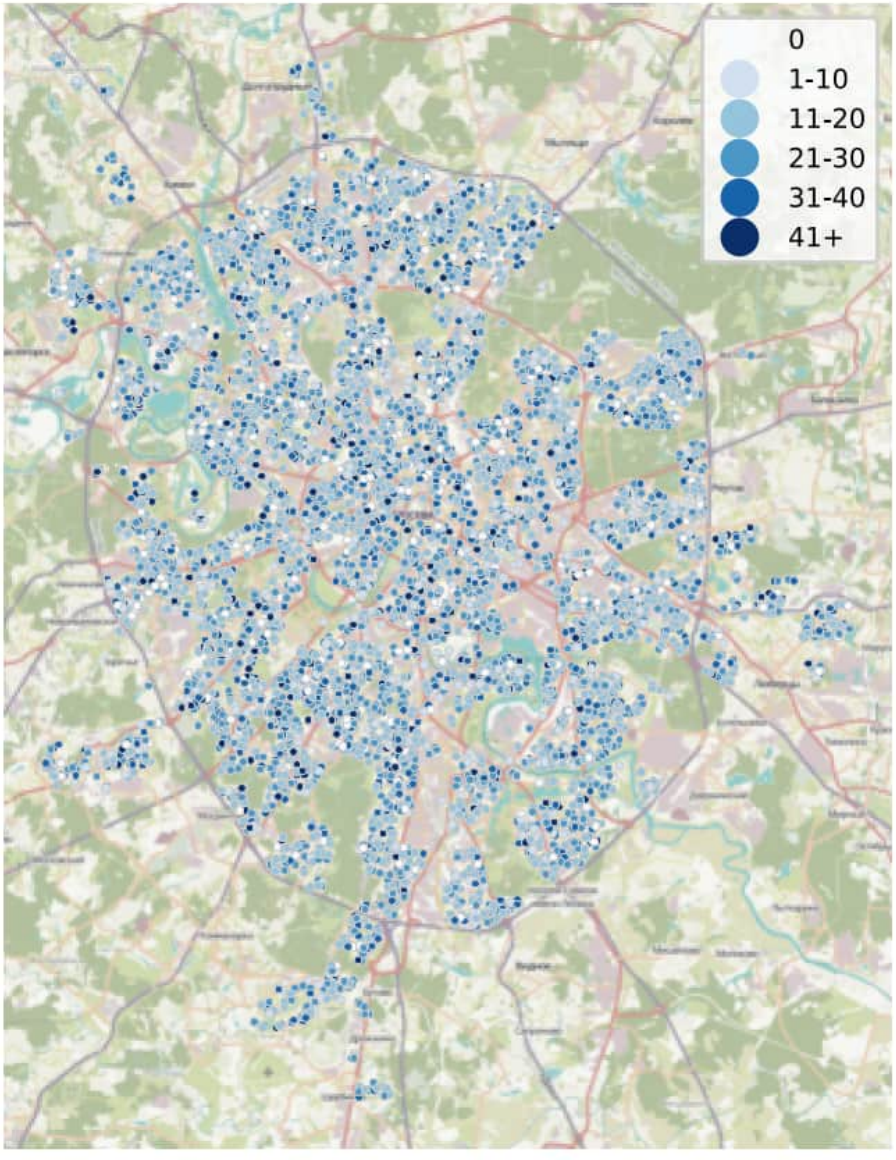
Notes: Ads posted in `cian.ru` between July 1st and August 14h, 2025. $N = 17{,}249/11{,}772$.

Data Collection Results

Long-term Rentals



Secondary Sales



First Approach: Querying a Multimodal LLM

- State-of-the-art LLMs offer the possibility of submitting queries that include images and other media
- The API is model-specific but typically follows a role-based structure
  - System Prompt: general instructions defining the type of queries the model might receive and the expected answer
  - Output Format Constraints (e.g. JSON schema)
  - Human Messages: text and other media

Scoring Real Estate Properties

- Querying a LLM can be costly and time-consuming
- One query per property, appending (as many as) ten photos picked at random
- Batches of 20–25 properties per request
- One retry for every unsuccessful submission

Querying Llama4 for Photo Ratings

System Prompt

```
You are a helpful assistant that can look at pictures of real
estate and provide accurate descriptions and analysis.  Answer
the user's questions regarding the attached images as clearly
and accurately as possible.
```

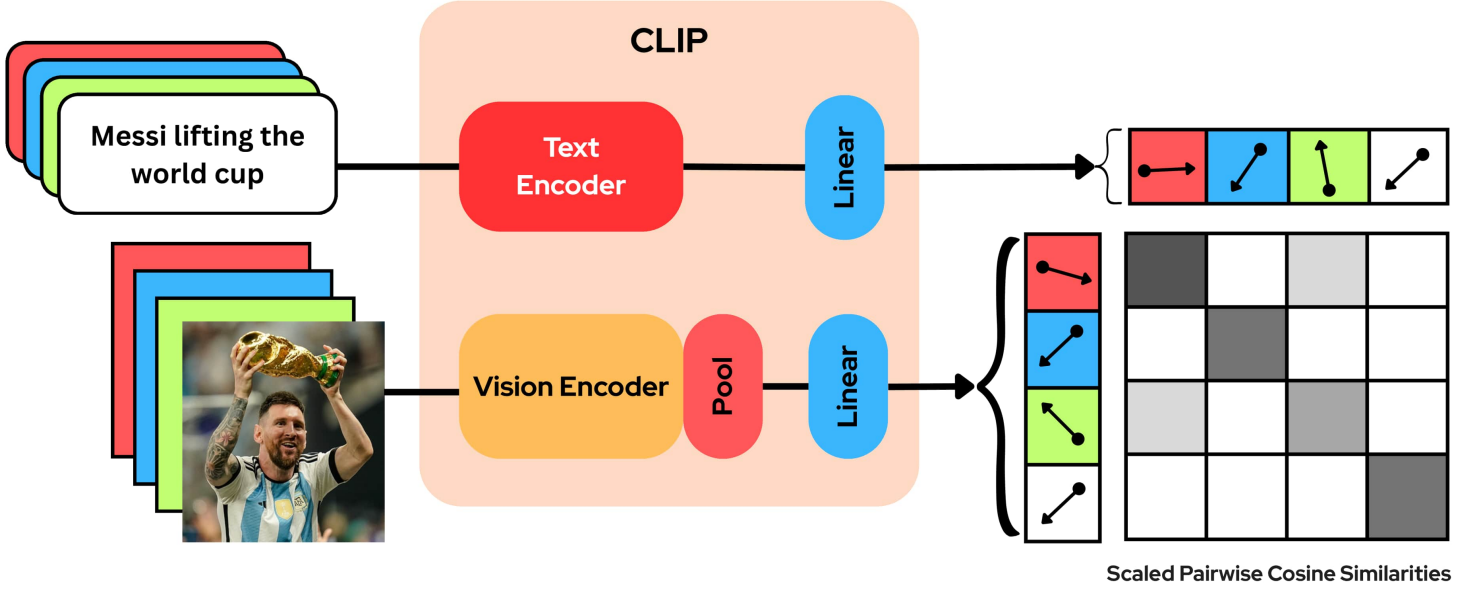
User Query

```
Rate how luxurious and upscale this apartment looks based on the
pictures.  Give a score from 0 to 10 to each photo.  Answer with
a list of scores (e.g.:  '[4,3,5,7,2,0,9]').  Omit any explanation
or additional information.

 Photo 1 URL                ................                Photo 10 URL
```

Second Approach: CLIP

- Contrastive Language-Image Pre-training (Radford et al., 2021)
- The model learns to encode text **and** images into vector representations
- The representations are trained to become similar when the text and image match, and dissimilar otherwise



CLIP Loss Function

- The vision and text encoders produce a batch of normalized embeddings:

$$\boldsymbol{E}_{img}, \boldsymbol{E}_{txt} \in \mathbb{R}^{n,d}$$

- The symmetric contrastive cross-entropy loss is calculated as follows:

$$\begin{aligned}
\boldsymbol{S} &= \boldsymbol{E}_{img} \cdot \boldsymbol{E}'_{txt} \\
\boldsymbol{L} &= \boldsymbol{S} * \tau \\
\text{loss}_{img} &= \text{colXentropy}(\boldsymbol{L}, [1, \ldots, n]) \\
\text{loss}_{txt} &= \text{rowXentropy}(\boldsymbol{L}, [1, \ldots, n]) \\
\text{loss} &= (\text{loss}_{img} + \text{loss}_{txt})/2
\end{aligned}$$

CLIP Inference: real estate photos

1. Submit each photo together with three alternative text descriptions
   - $+$ This is a photo of a *luxurious* apartment.
   - $\sim$ This is a photo of an *ordinary* apartment.
   - $-$ This is a photo of a *dilapidated* apartment.
2. Use the text and image embeddings to obtain similarity scores

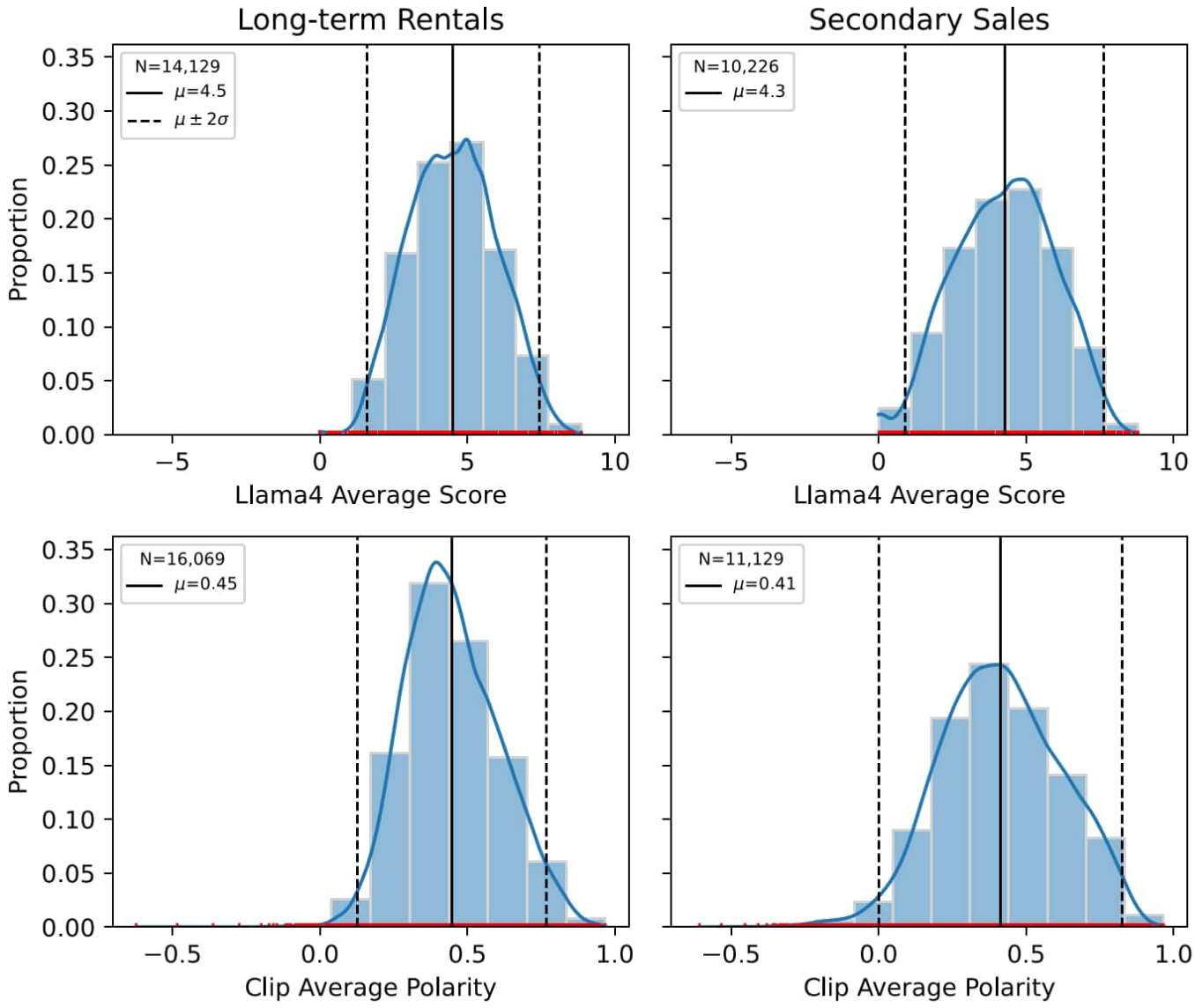
$$S(j) = \boldsymbol{E}_i \cdot \boldsymbol{E}_j, \quad j \in \{+, \sim, -\}$$

3. Use model temperature to get logits: $L(j) = S(j) * \tau$
4. Transform logits into probabilities

$$P(j) = \frac{\exp(L(j))}{\sum_{h \in \{+,-,\sim\}} \exp(L(h))}, \quad j \in \{+, \sim, -\}$$

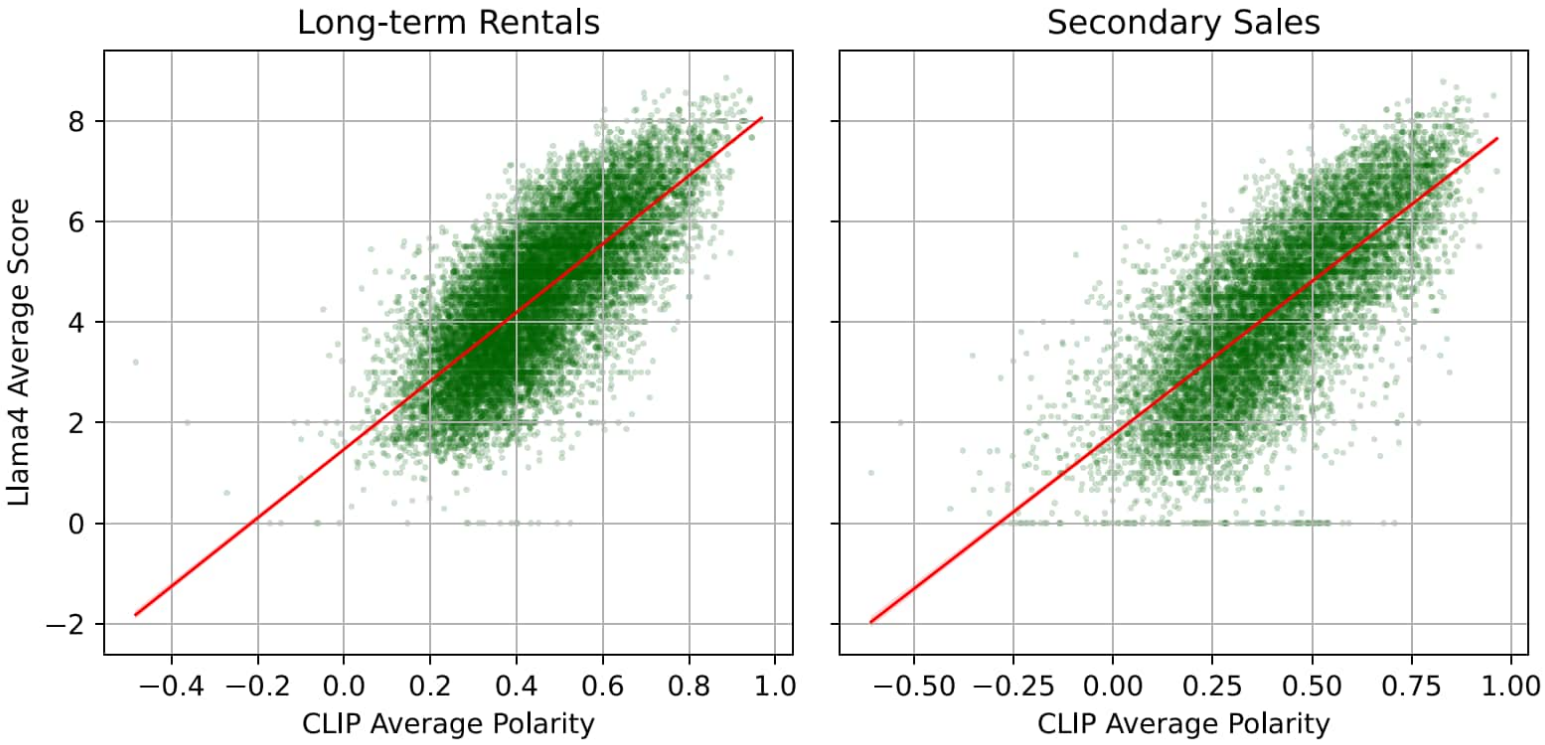
5. $\text{polarity}_{CLIP} = P(+) - P(-)$

Comparing Approaches



Note: Based on $N = 16{,}578/11{,}213$ ads with at least one photo.

Comparing Approaches

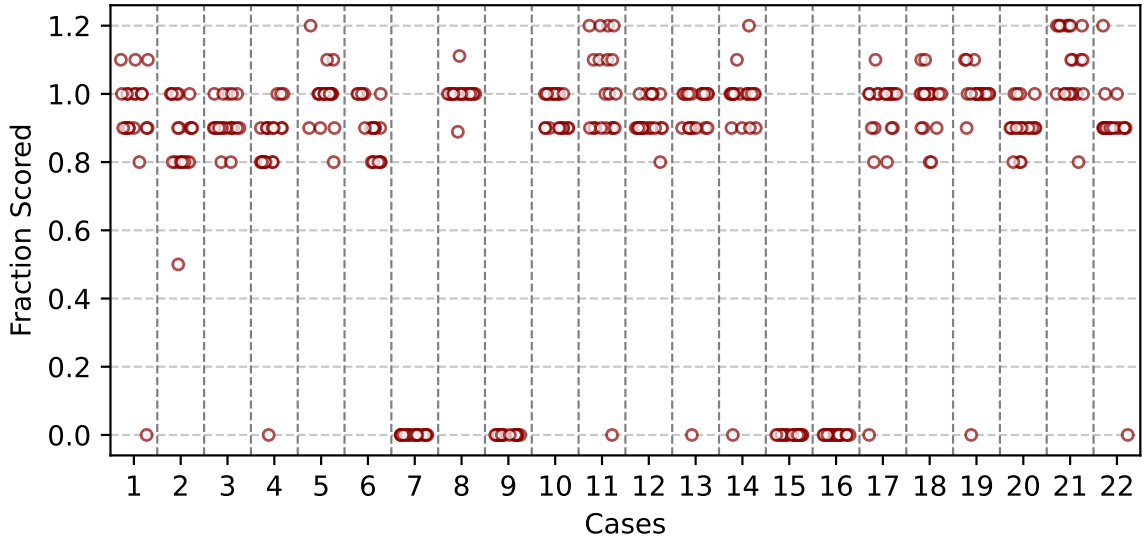


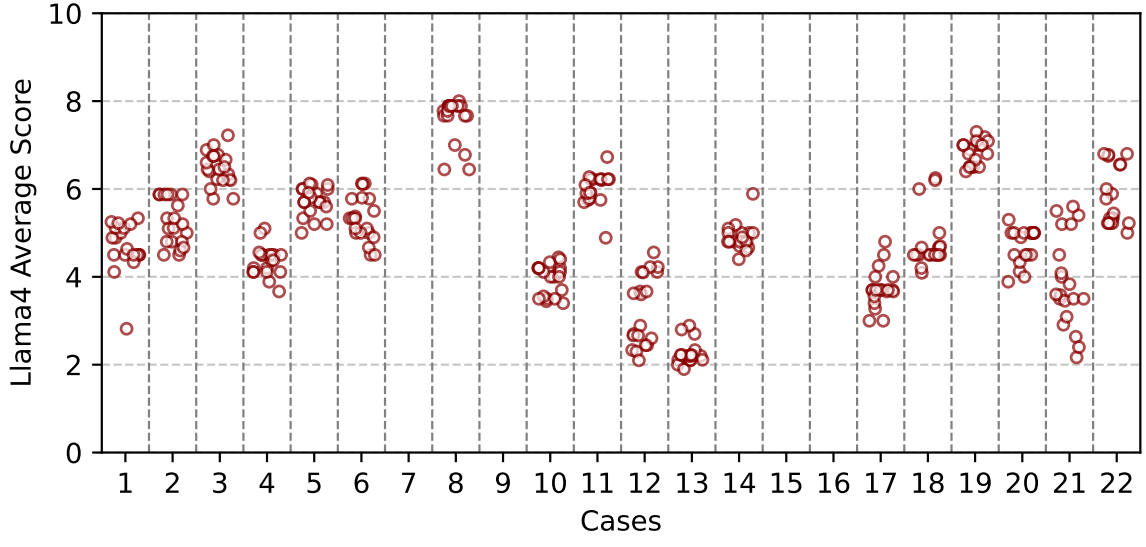
Notes: $N = 14{,}129/10{,}226$.

Replication Study

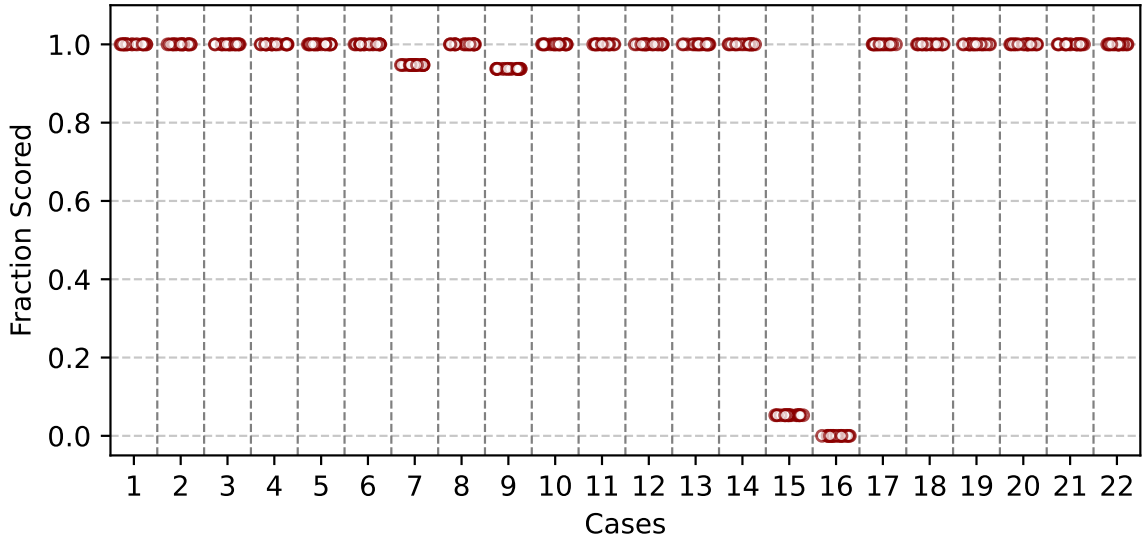
- Batch of 22 randomly selected properties offered for long-term rental
- Twenty replications with initial reshuffling of the batch
- Two outcomes for each property:
  1. Fraction of scored photos
  2. Average score

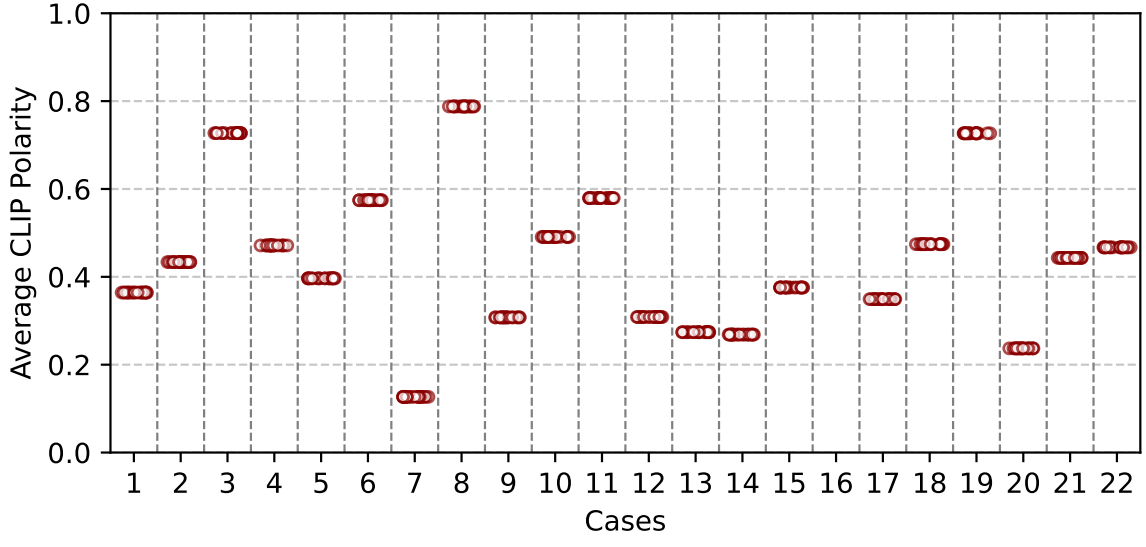
Llama4 Scoring Replication Results



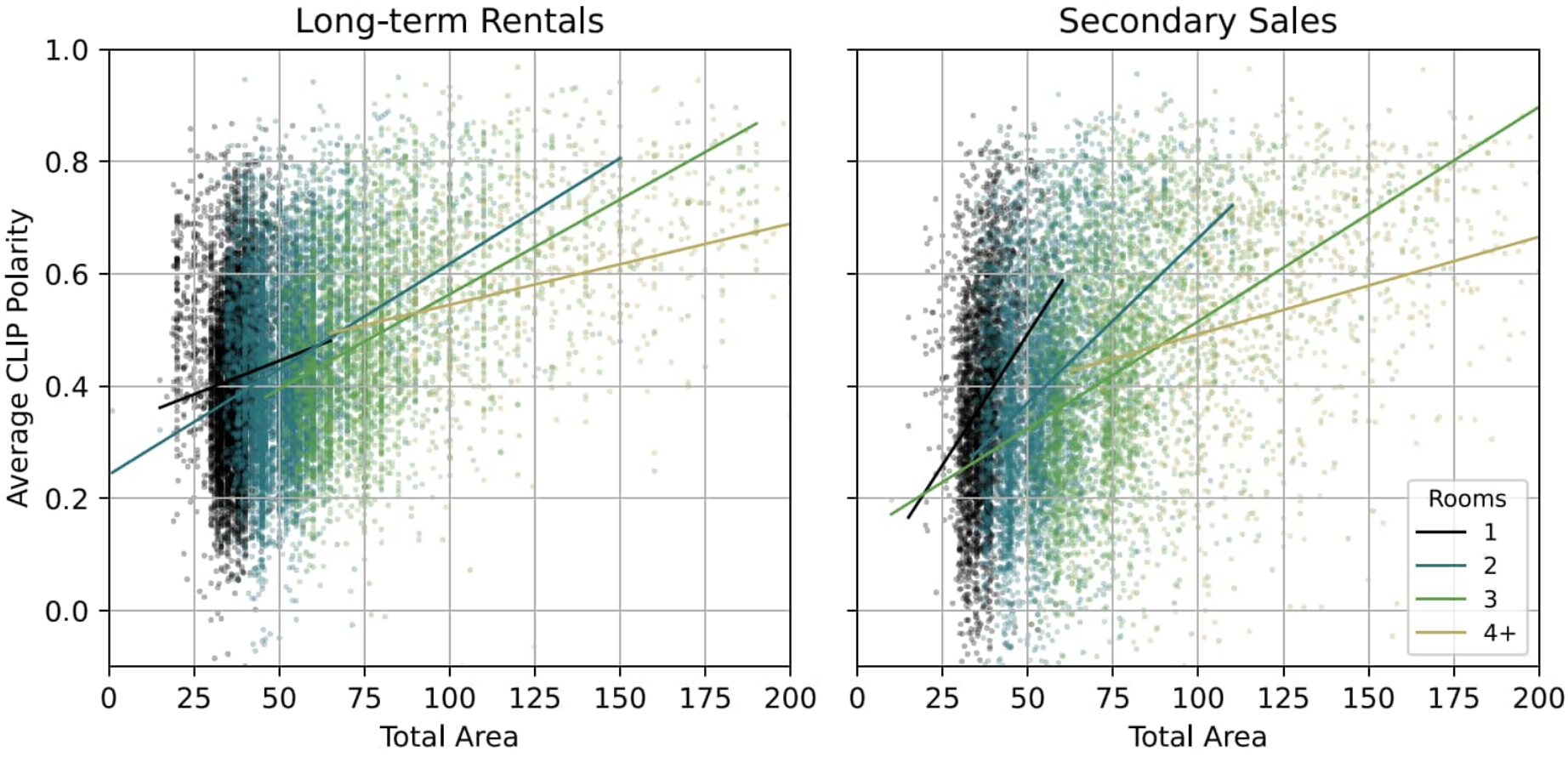


CLIP Scoring Replication Results

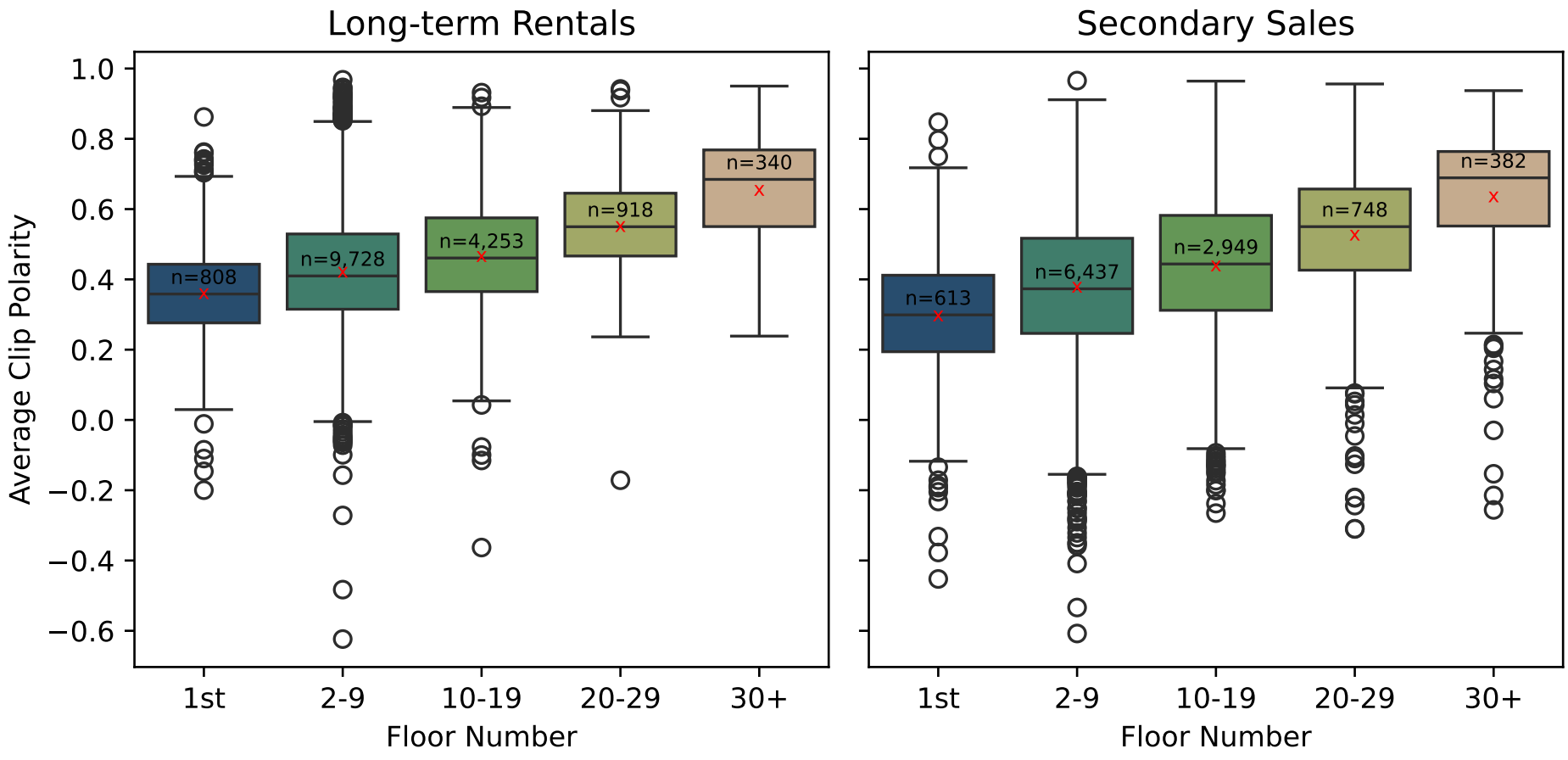




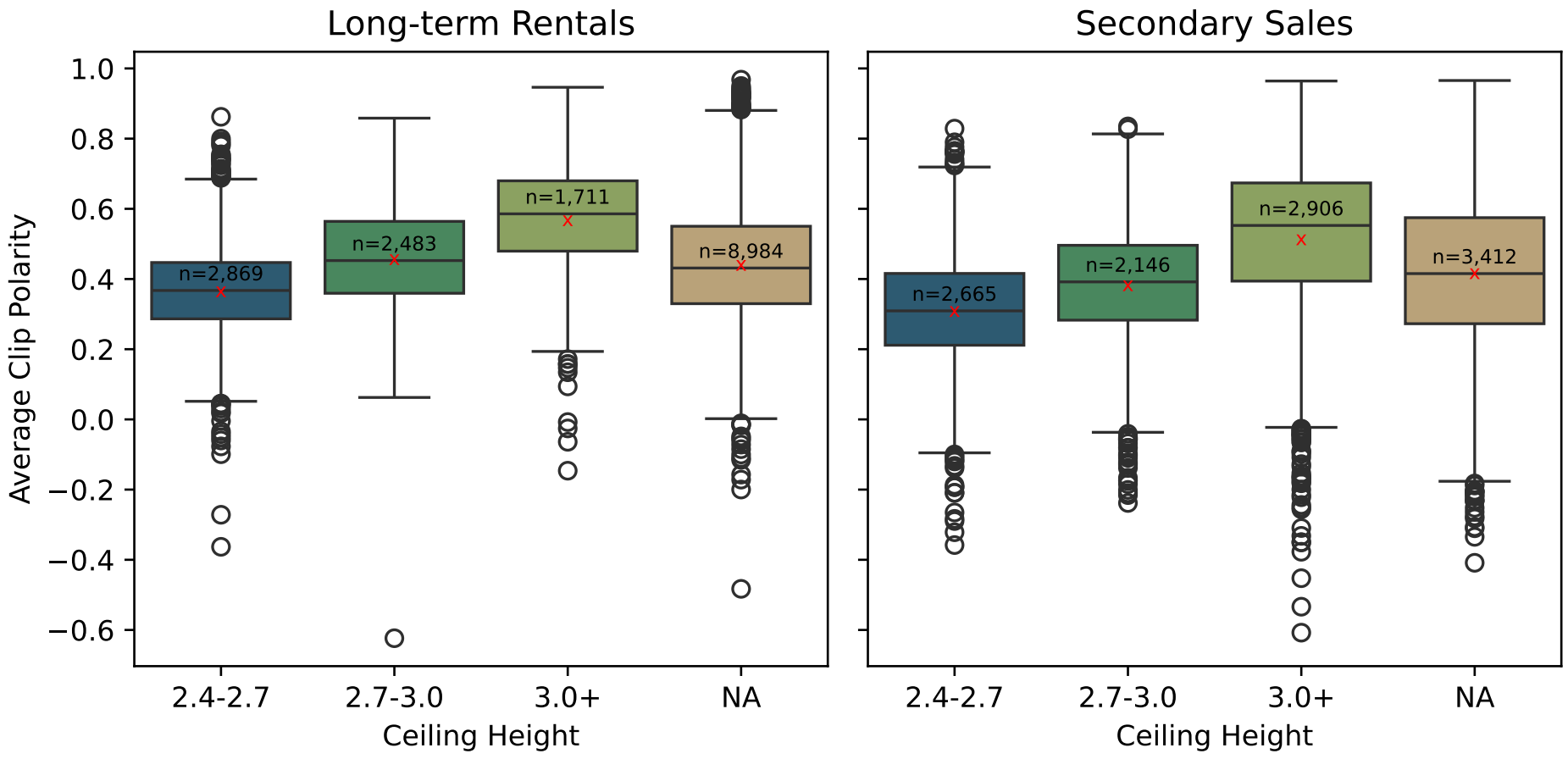
CLIP Polarity and Housing Characteristics



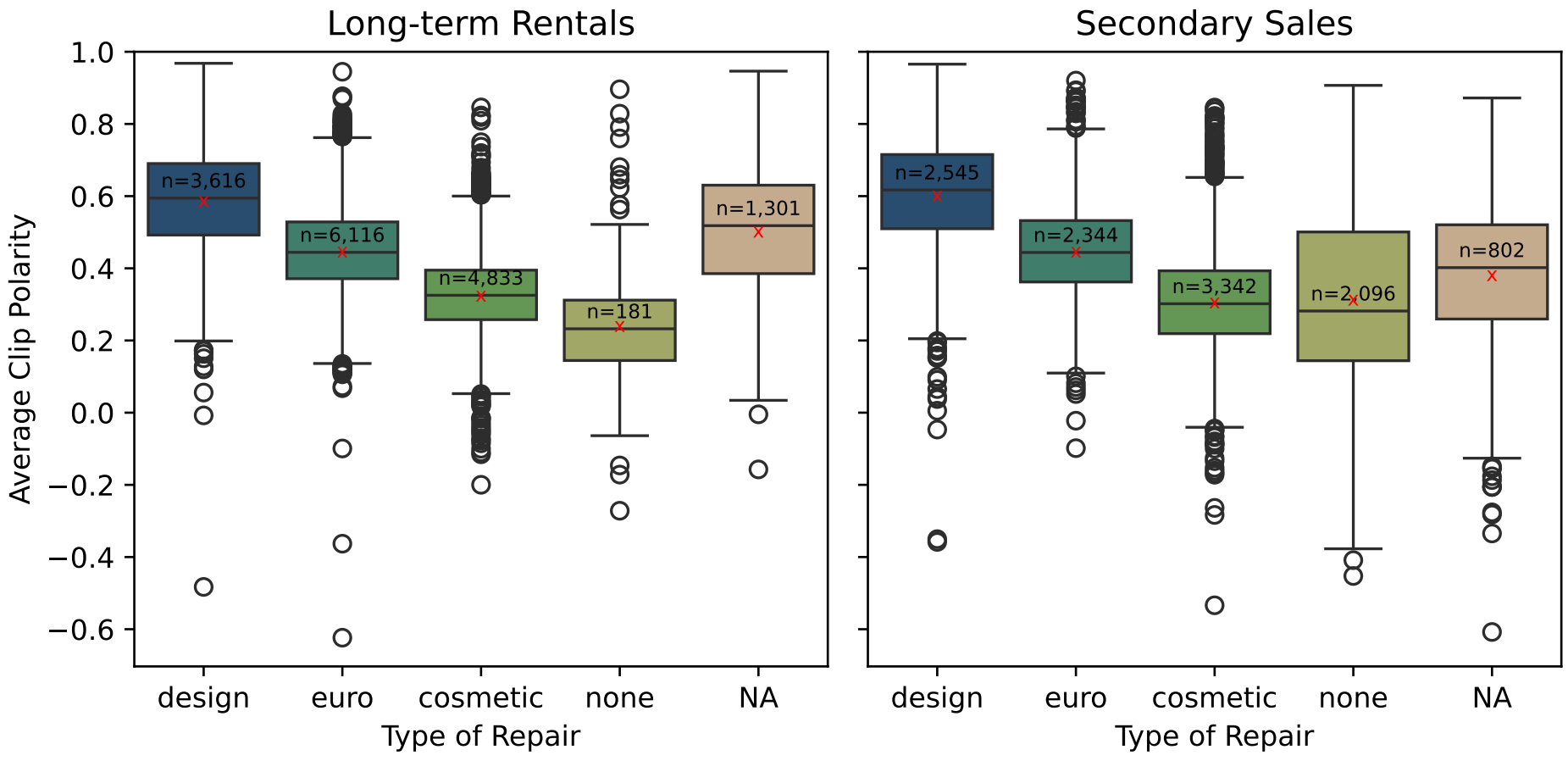
CLIP Polarity and Housing Characteristics



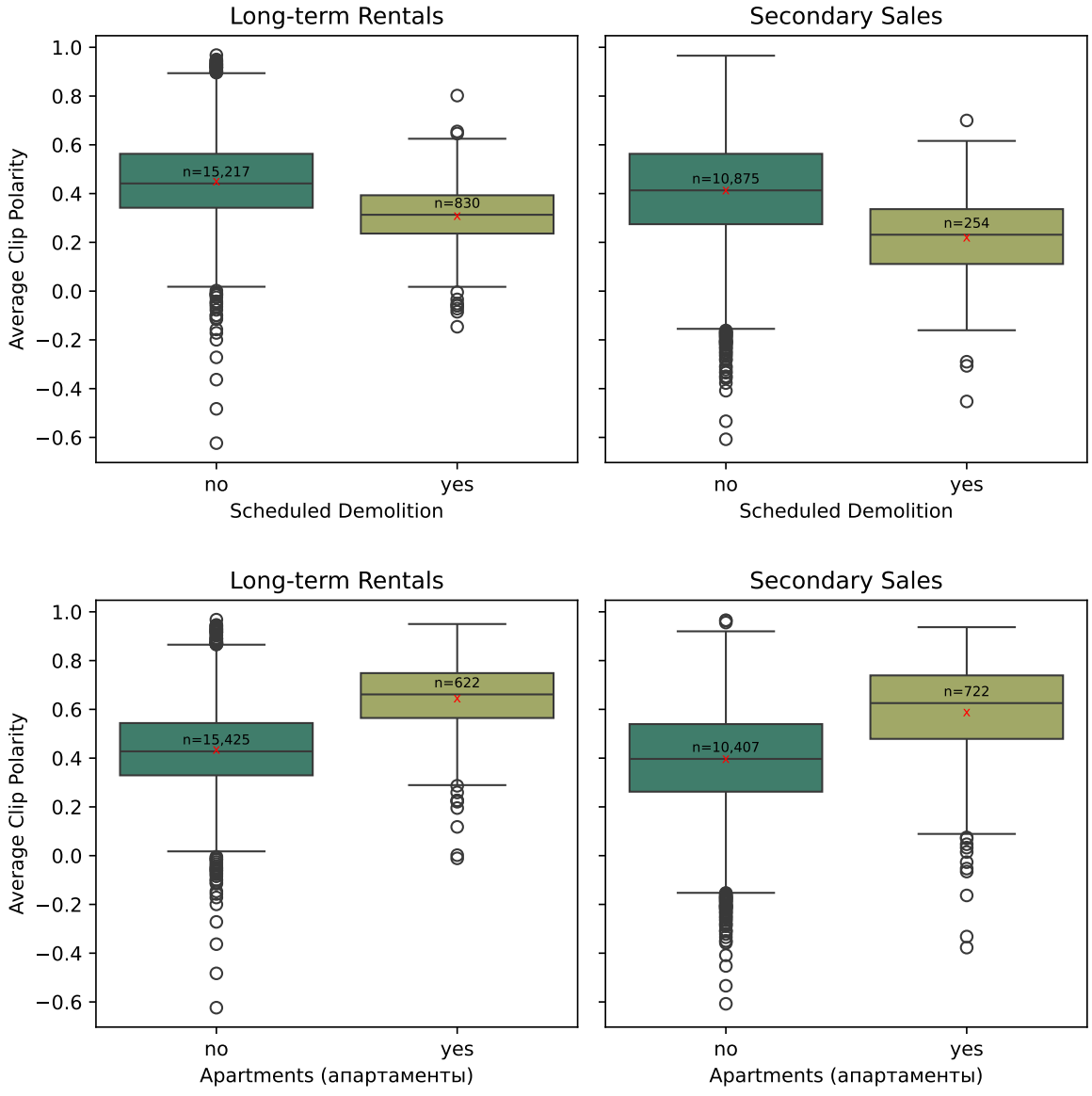
CLIP Polarity and Housing Characteristics



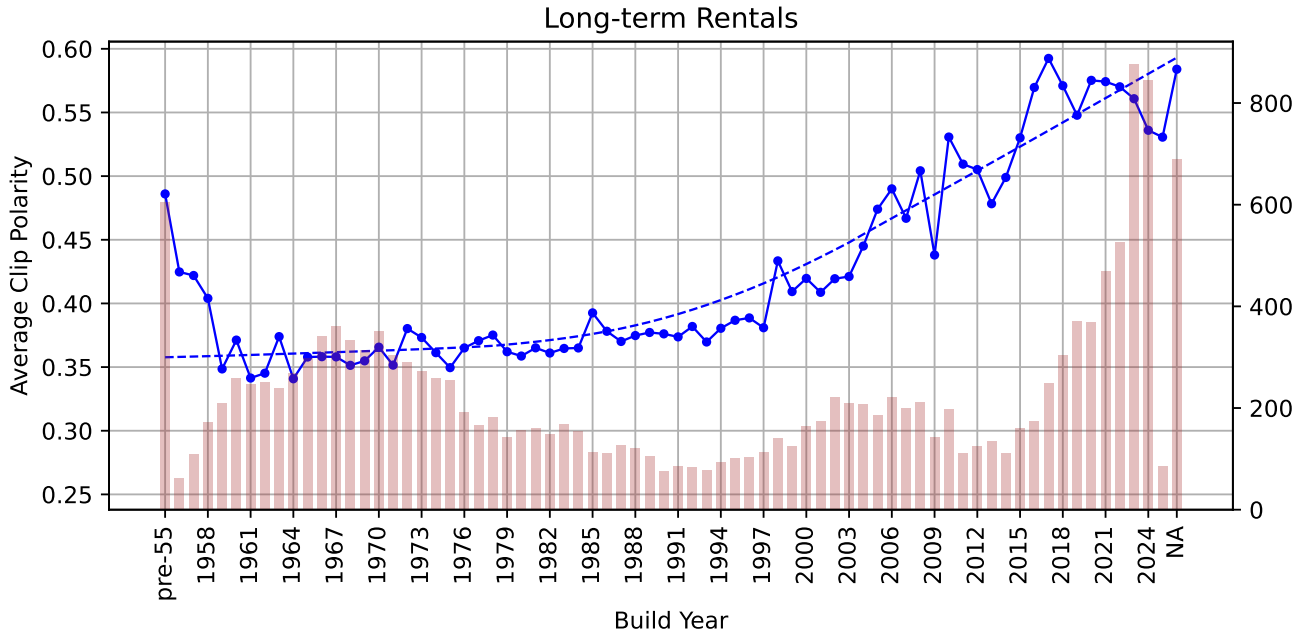
CLIP Polarity and Housing Characteristics

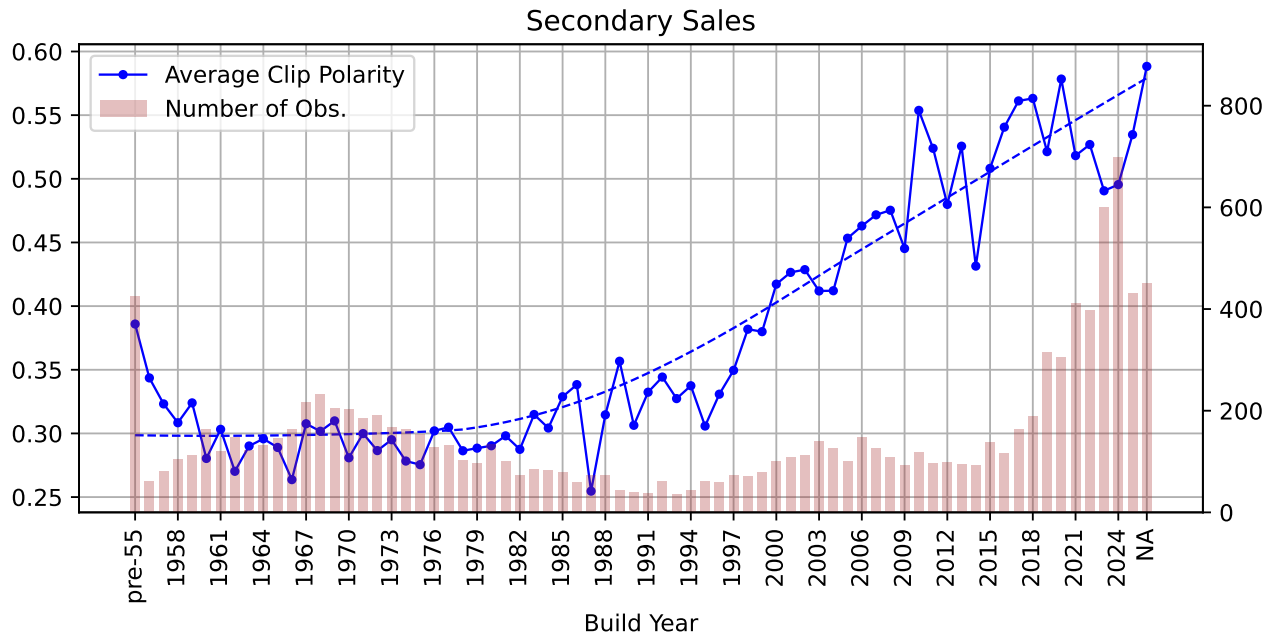


CLIP Polarity and Housing Characteristics

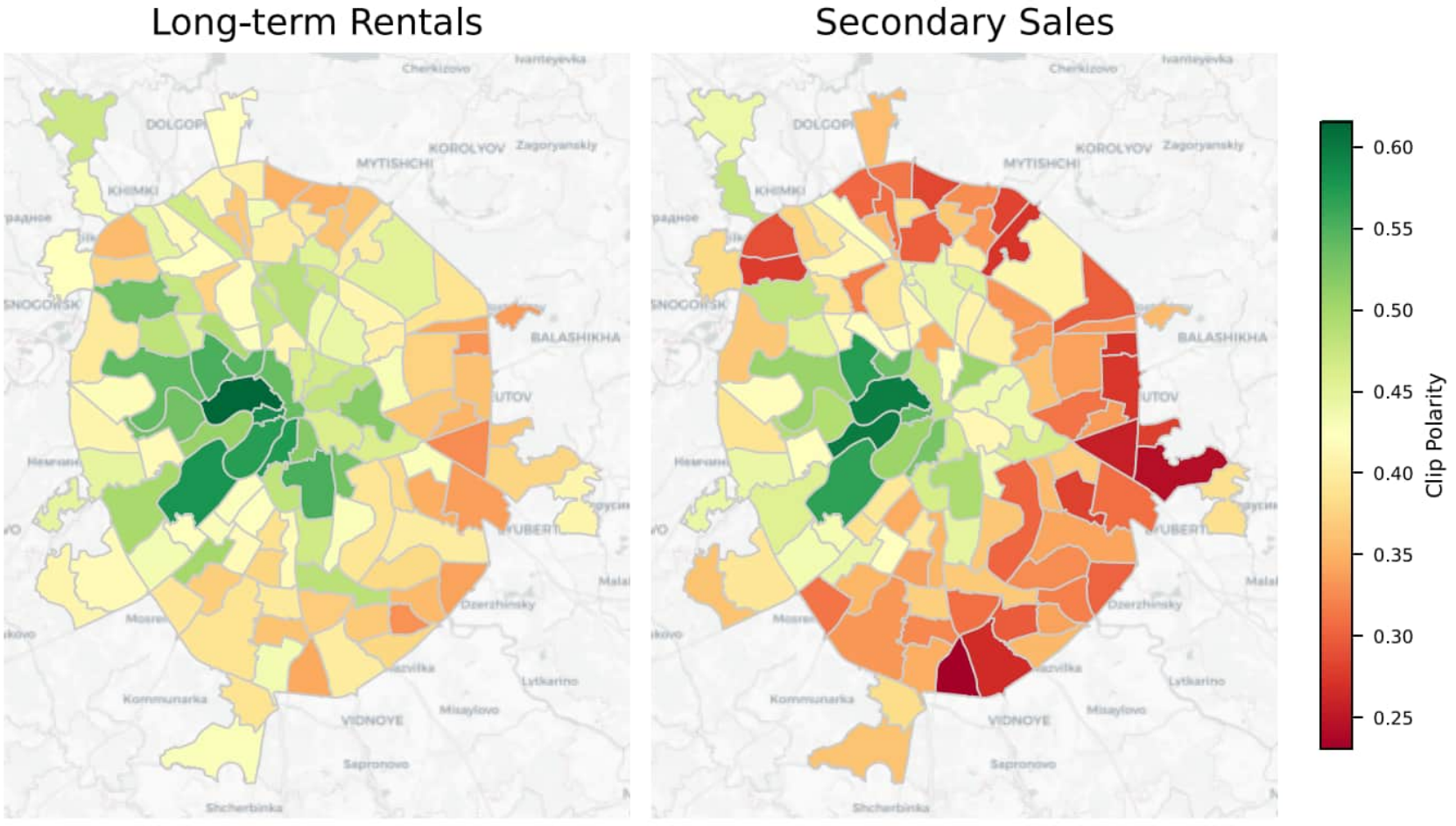


CLIP Polarity and Housing Characteristics





CLIP Polarity Choropleth



Hedonic Model

- Relates property attributes and prices (Rosen, 1974)
  - Unit-level: bedrooms, bathrooms, square meterage, ceiling height
  - Building-level: age, construction material
  - Geographic attributes: distance to amenities, transportation access
  - District-level: school quality, crime rates, etc.
- We explore whether CLIP contains valuable information toward pricing the property

Hedonic Model Specifications

- The most common approach in the literature specifies a log-linear model

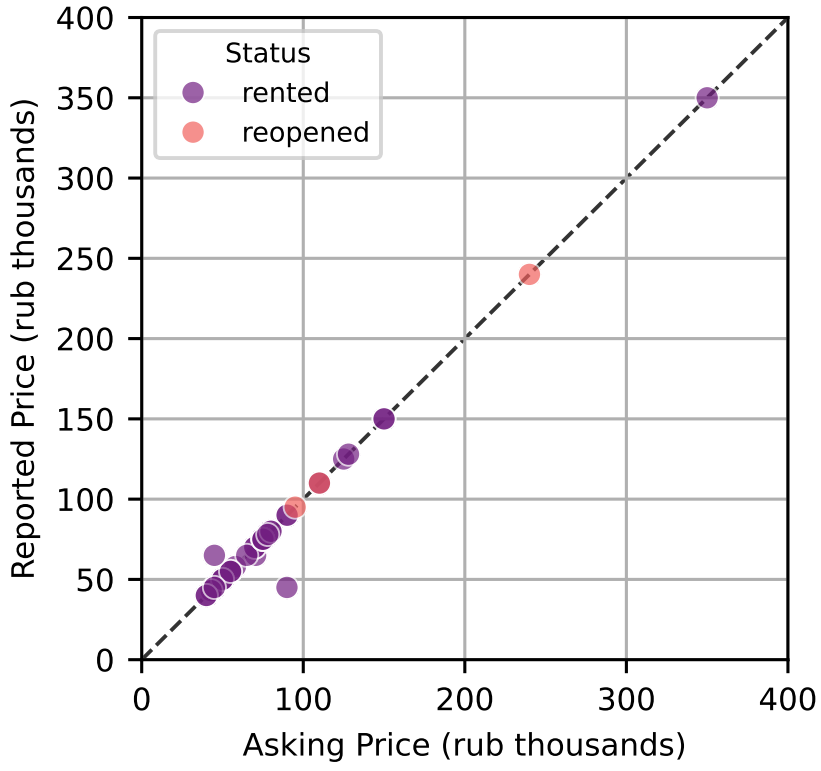
$$\log p_i = \alpha + \beta \cdot \text{CLIP}_i + \boldsymbol{x}_i' \boldsymbol{\delta} + \varepsilon_i$$

- We also train a histogram-based gradient boosting predictor
- Both predictors are estimated on a training sample (80%; stratified by district)
- We look at prediction performance on both the training and the test sample
- We compute the *importance* of CLIP polarity
  - Shapely value

Asking Prices v. Transaction Prices

- We observe asking prices and not transaction prices
- We conducted a small survey ($N = 173$) to try to gauge potential measurement error

Status	N	% positive
no answer	109	
rented	51	79.7
reopened	10	15.6
quit	3	4.7



Hedonic Model Performance

Table: Long-term Rentals

	(1a)	(1b)	(2a)	(2b)	(3a)	(3b)	(4a)	(4b)
CLIP	✗	✓	✗	✓	✗	✓	✗	✓
Coefficient Estimates								
Avg. CLIP Polarity		1.037***		0.855***		0.686***		0.597***
(HC3 S.E.)		(0.022)		(0.022)		(0.019)		(0.018)
In-sample								
Adj-R2	0.713	0.776	0.771	0.808	0.840	0.863	0.870	0.888
RMSE	0.270	0.239	0.241	0.220	0.201	0.186	0.181	0.168
Out-of-sample								
Adj-R2	0.741	0.793	0.786	0.818	0.852	0.871	0.873	0.887
RMSE	0.261	0.233	0.237	0.219	0.197	0.184	0.179	0.169
Other Included Features								
Unit	✓	✓	✓	✓	✓	✓	✓	✓
Building	✗	✗	✓	✓	✓	✓	✓	✓
Geography	✗	✗	✗	✗	✓	✓	✓	✓
District FE	✗	✗	✗	✗	✗	✗	✓	✓

Notes: train set N=12,725; test set N=3,182.

Hedonic Model Performance

Table: Secondary Sales

	(1a)	(1b)	(2a)	(2b)	(3a)	(3b)	(4a)	(4b)
CLIP	✗	✓	✗	✓	✗	✓	✗	✓
Coefficient Estimate								
Avg. CLIP polarity		0.791***		0.619***		0.458***		0.363***
(HC3 S.E.)		(0.024)		(0.025)		(0.022)		(0.020)
In-sample Performance								
Adj-R2	0.749	0.788	0.805	0.825	0.870	0.881	0.905	0.912
RMSE	0.315	0.290	0.278	0.263	0.227	0.217	0.192	0.185
Out-of-sample Performance								
Adj-R2	0.748	0.791	0.798	0.821	0.875	0.886	0.903	0.910
RMSE	0.311	0.282	0.277	0.261	0.218	0.208	0.187	0.179
Other Included Features								
Unit	✓	✓	✓	✓	✓	✓	✓	✓
Building	✗	✗	✓	✓	✓	✓	✓	✓
Geography	✗	✗	✗	✗	✓	✓	✓	✓
District FE	✗	✗	✗	✗	✗	✗	✓	✓

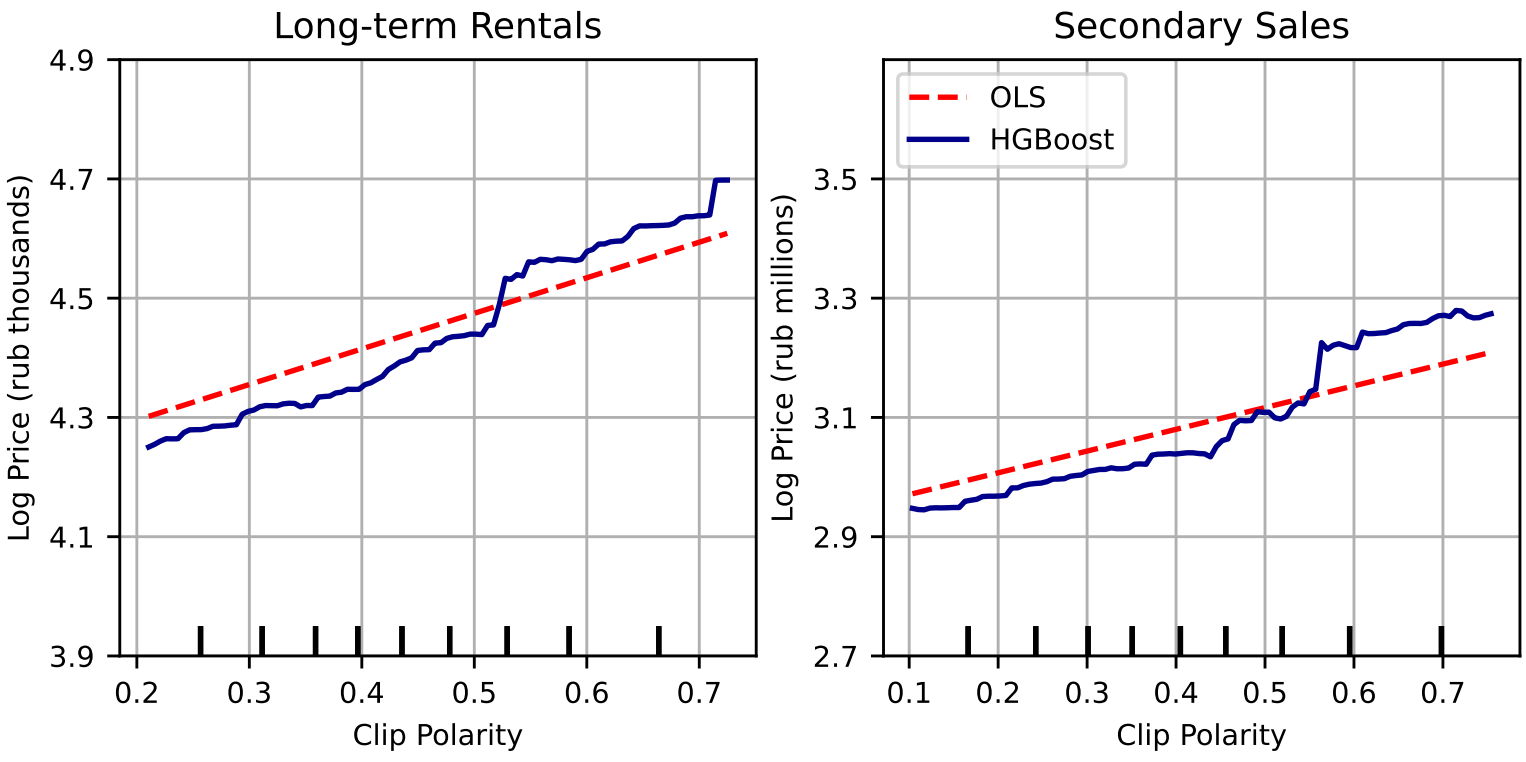
Notes: train set N=8,812; test set N=2,204.

Gradient Boosting

- An ensemble technique that builds a strong predictive model by sequentially combining multiple “weak” learners
- Each new tree specifically designed to correct the errors made by all previous trees
- Key hyper-parameters (N_{trees}, lr, $depth$) chosen by randomized CV grid search

Gradient Boosting

Figure: Partial Dependence



Gradient Boosting

Table: Performance Measures

	Long-term Rentals		Secondary Sales	
CLIP	✗	✓	✗	✓
In-sample Performance				
Adj-R2	0.921	0.933	0.958	0.960
RMSE	0.141	0.130	0.128	0.126
Out-of-sample Performance				
Adj-R2	0.888	0.900	0.925	0.928
RMSE	0.172	0.163	0.169	0.166

Notes: train set N=8,812; test set N=2,204. Unit, building, geographic, and district level features included.

Shapley Values

- It is possible to decompose the contribution of each feature in a prediction problem
- Axioms: efficiency, symmetry, additivity, null player (Shapley, 1953)
- Unique solution

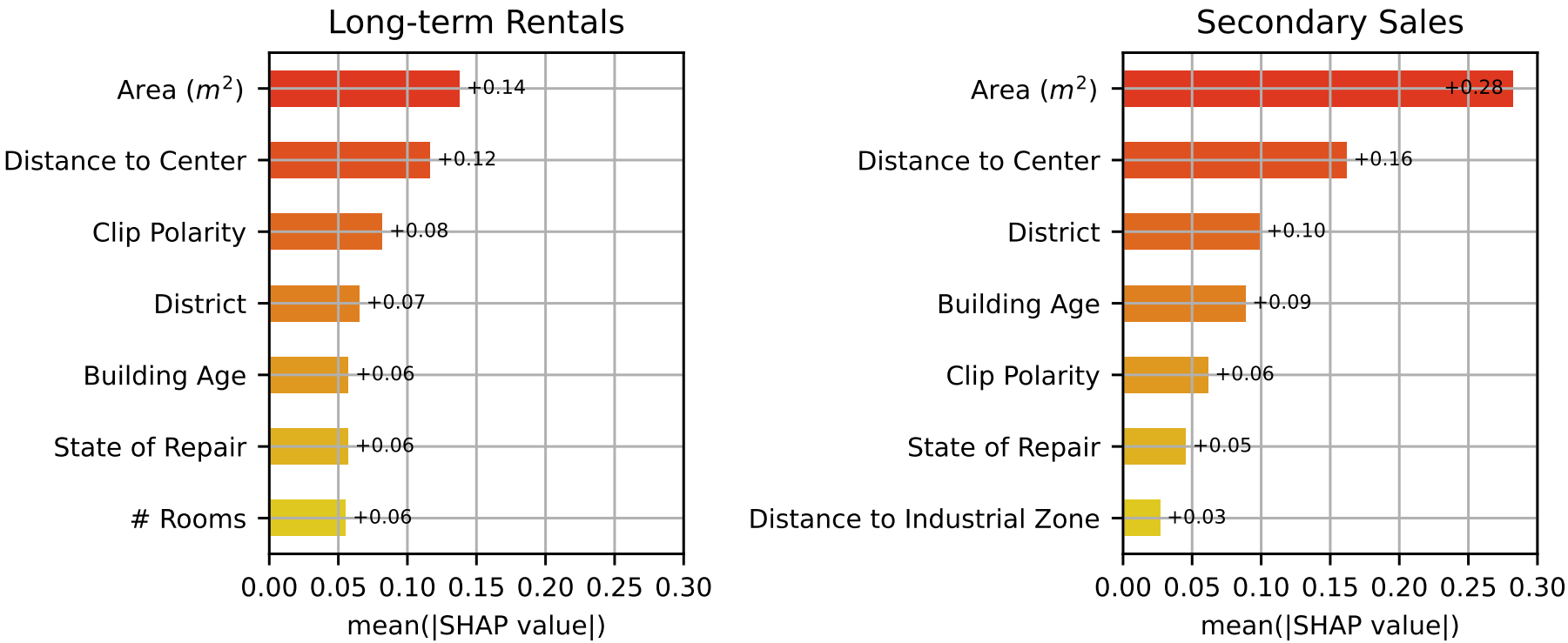
$$N = \{1, 2, \ldots, n\}$$

$$\varphi_j = \frac{1}{n} \sum_{S \subseteq N \setminus \{j\}} \binom{n-1}{|S|}^{-1} [v(S \cup \{j\}) - v(S)]$$

The Shapley value (φ_j) is the weighted average of j's marginal contributions across all possible coalitions.

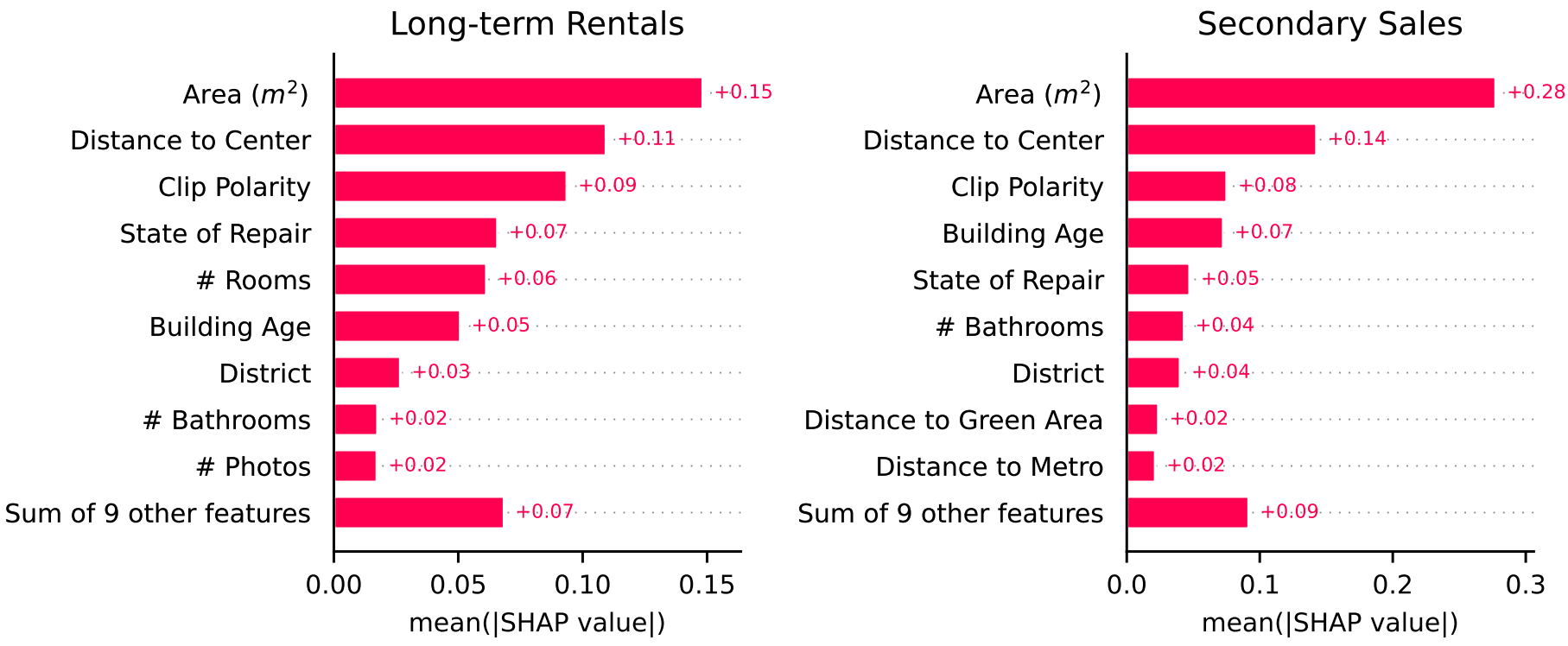
Shapely Values

Figure: Linear Regression



Shapely Values

Figure: Gradient Boosting



Time on the Market

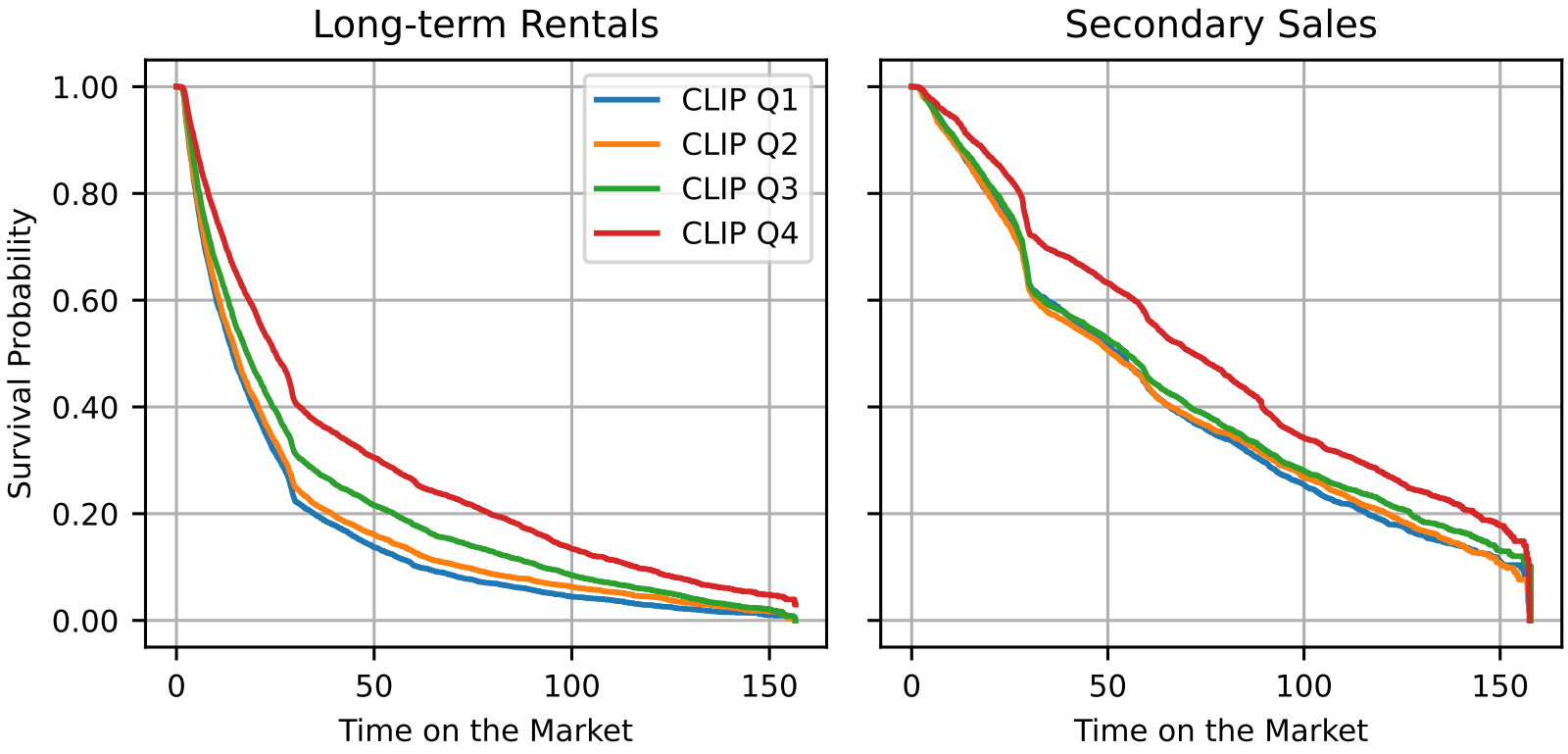
- We follow ads for up to 157 days
- A small fraction ($\sim 0.1\%$) of properties are lost to follow-up during the analysis period

Table: Time on the Market

	Count	Mean	Std.Dev.	Min	Median	Max
Long-term Rentals						
Closed	15,245	27.5	30.0	1.2	16.4	156.6
Censored	663	117.8	32.7	2.4	123.6	156.6
Secondary Sales						
Closed	8,981	51.3	37.1	1.3	40.6	157.8
Censored	2,035	126.6	21.6	3.6	128.0	157.4

CLIP and Time on the Market

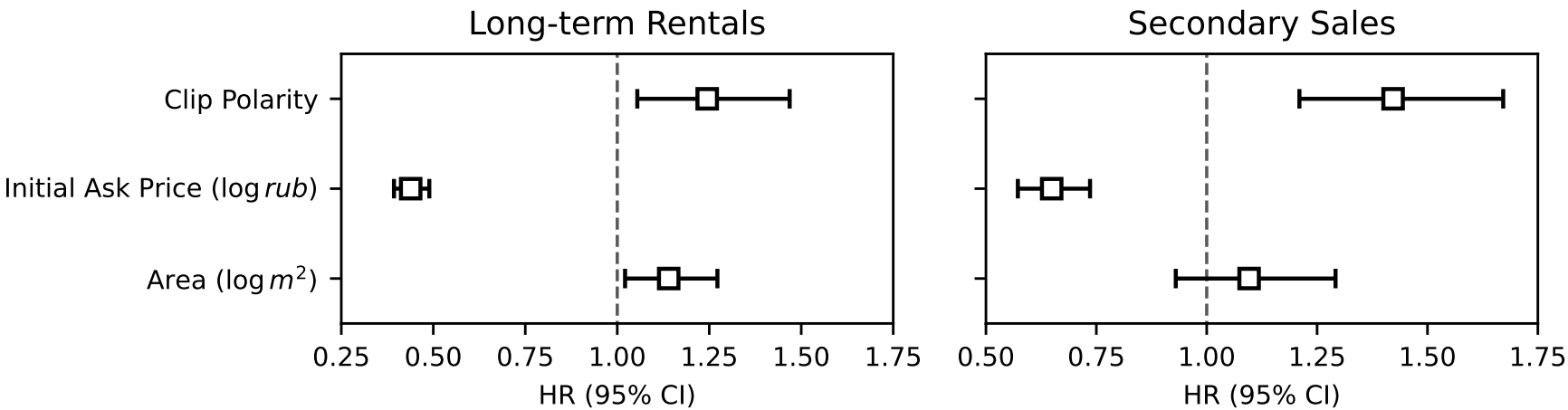
Figure: Kaplan-Meier Curves by CLIP Polarity Quartile



Proportional Hazard Model

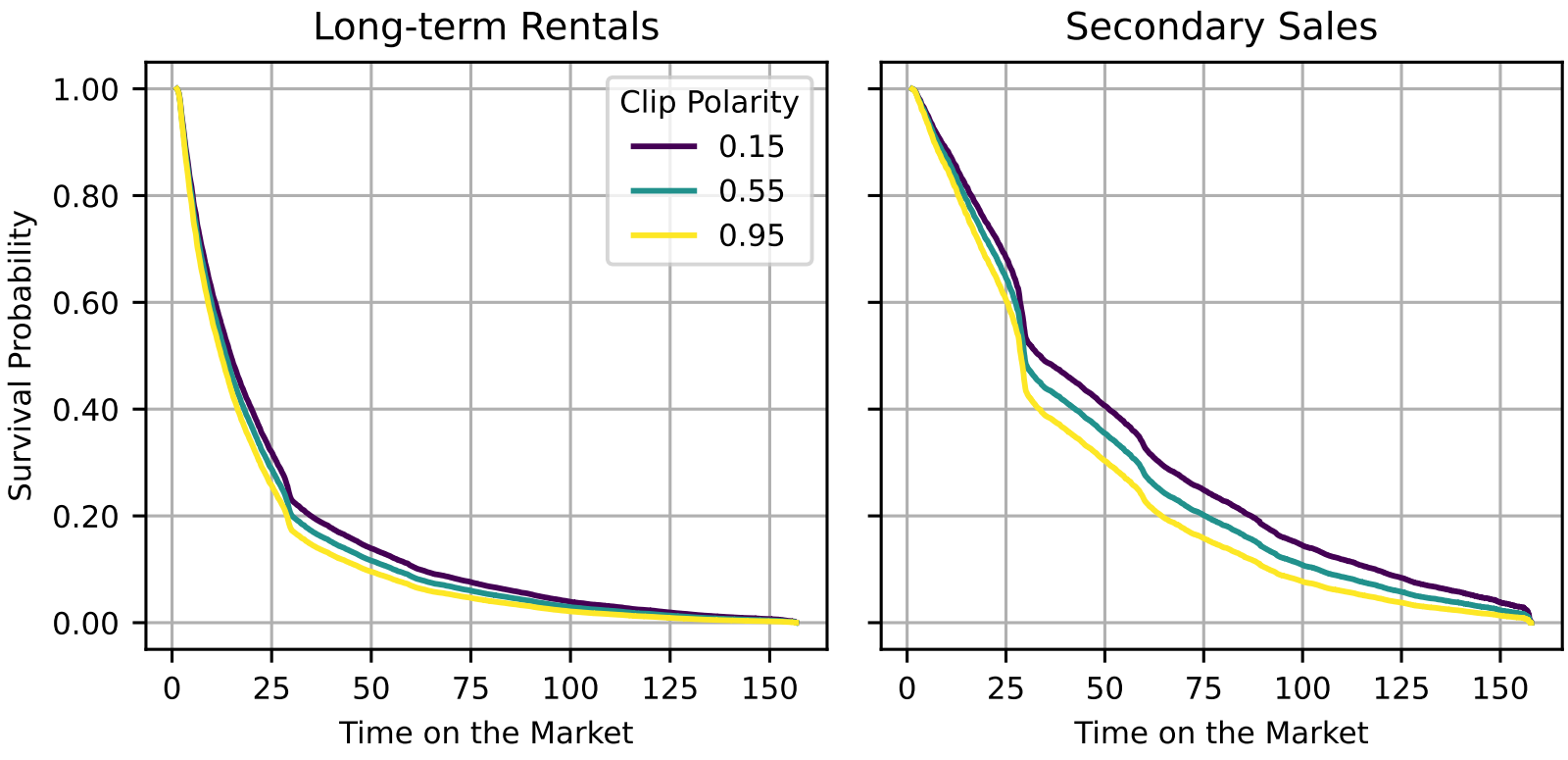
$$h(t \mid \boldsymbol{x}) = h_0(t) \exp\left(\boldsymbol{x}'\boldsymbol{\theta}\right)$$

Figure: Hazard Ratios for Selected Features



Proportional Hazard Model

Figure: Partial Effects



Conclusions

- CLIP is a new approach to extract information from images
- It is fully reproducible, relatively efficient computationally, and free

Conclusions

- Predictably, better-looking real estate tends to be located in upscale neighborhoods
- The CLIP-score is predictive of prices and time on the market

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